
AGENTIC RETRIEVAL-AUGMENTED GENERATION: A SURVEY ON AGENTIC RAG

Aditi Singh

Department of Computer Science
Cleveland State University
Cleveland, OH, USA
a.singh22@csuohio.edu

Abul Ehtesham

Kent State University
Kent, OH, USA
aehtesha@kent.edu

Saket Kumar

Khoury College of Computer Science
Northeastern University
Boston, MA, USA
kumar.sak@northeastern.edu

Tala Talaei Khoei

Khoury College of Computer Science
Roux Institute at Northeastern University
Portland, ME, USA
t.talaeikhoei@northeastern.edu

Athanasios V. Vasilakos

Center for AI Research
University of Agder (UiA)
Jon Lilletuns vei 9, 4879 Grimstad, Norway
th.vasilakos@gmail.com

ABSTRACT

Large Language Models (LLMs) have significantly advanced artificial intelligence by enabling human-like text generation and natural language understanding. However, their reliance on static training data limits their ability to respond to dynamic, real-time queries, resulting in outdated or inaccurate outputs. Retrieval-Augmented Generation (RAG) has emerged as a solution, enhancing LLMs by integrating real-time data retrieval to provide contextually relevant and up-to-date responses. Despite its promise, traditional RAG systems are constrained by static workflows and lack the adaptability required for multi-step reasoning and complex task management. Agentic Retrieval-Augmented Generation (Agentic RAG) transcends these limitations by embedding autonomous AI agents into the RAG pipeline. These agents leverage agentic design patterns reflection, planning, tool use, and multi-agent collaboration to dynamically manage retrieval strategies, iteratively refine contextual understanding, and adapt workflows through clearly defined operational structures ranging from sequential steps to adaptive collaboration. This integration enables Agentic RAG systems to deliver unparalleled flexibility, scalability, and context-awareness across diverse applications. This paper presents an analytical survey of Agentic RAG¹ systems. It traces the evolution of RAG paradigms, introduces a principled taxonomy of Agentic RAG architectures based on agent cardinality, control structure, autonomy, and knowledge representation, and provides a comparative analysis of design trade-offs across existing frameworks. The survey further examines real-world applications in domains including healthcare, finance, education, and enterprise document processing, and distills practical lessons learned for system designers and practitioners. Finally, it identifies key open research challenges related to evaluation, coordination, memory management, efficiency, and governance, outlining directions for future research.

Keywords Large Language Models (LLMs) · Artificial Intelligence (AI) · Natural Language Understanding · Retrieval-Augmented Generation (RAG) · Agentic RAG · Autonomous AI Agents · Reflection · Planning · Tool Use · Multi-Agent Collaboration · Agentic Patterns · Contextual Understanding · Dynamic Adaptability · Scalability · Real-Time Data Retrieval · Taxonomy of Agentic RAG · Healthcare Applications · Finance Applications · Educational Applications · Ethical AI Decision-Making · Performance Optimization · Multi-Step Reasoning

¹*GitHub link:* <https://github.com/asinghcsu/AgenticRAG-Survey>